

KPIs, variables, grain, chart critique, bounded data stories

Quantitative Methods for Business Analysis. Complete explanations, worked examples, practice and transfer cases for every idea in the session.

≈ 59 min read

Download the PDF

Concise edition

1 · READ

Open a section, read its central idea, then its worked example.

2 · APPLY

Try the guided practice and the lab task linked in each section.

3 · CHECK

Answer the study check in the writing space without looking back.

Your study route

2.1

Decision Frame, KPIs, Targets, and Guardrails

5 topics · slides 9–19 · Part A · Part B

2.2

Reproducible KPI Definitions, Aggregation, and Center and Position

5 topics · slides 20–38 · Part C · Part D · Part L

2.3

Variables, Measurement Scales, Observations, and Grain

5 topics · slides 39–54 · Part E · Part F · Part K

2.4

Honest Charts, Bounded Stories, and Validation

5 topics · slides 55–70 · Part G · Part H · Part I · Part J · Part M · Part N

▼ What you will be able to do

- Translate a vague managerial request into a stakeholder, a decision, an action horizon, a KPI, a target, a guardrail and a bounded question.
- Distinguish a KPI, a management target and a guardrail, and explain why passing one does not answer the question asked by another.
- Calculate counts, proportions and target gaps, and report a gap between two rates in percentage points rather than as a percent change.

- State the unit, eligible population, time window, grain, numerator, denominator and direction of a KPI so another analyst can reproduce it.
- Combine weekly rates by summing counts before dividing, and explain when an average of rates gives the wrong answer.
- Interpret the mean, median and 90th percentile of late minutes as descriptions of center and position, and name what they cannot establish.

▼ Before you start

Read the PDF edition for study, annotation or printing; the HTML, notebook and Markdown editions are also available. A running kernel, software installation and an account connection are not required. Keep notes in the response spaces or in your own study document.

The formulas are complete teaching references. Worked results come from the session materials and the supplied file `orders_story.csv` ; where this toolkit calculates a value from that file, the calculation is stated beside it.

Before reading, write one idea you already understand and one question you want the session to answer.

Starting notes

My prior understanding and learning question:

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2.1 · Decision Frame, KPIs, Targets, and Guardrails

IN CLASS

Slides 9–19

Decision-First Analysis 10–11

KPI, Target, and Guardrail 12–14

MetroMeal's Bounded Question 15–16

Retrieval Check 18

PRACTISE IT IN THE LAB

- Part A: What decision can the KPI inform? · 3 min
- Part B: Is MetroMeal's latest week below target? · 3 min · SPSS Block H1: decision, KPI, target, and guardrail

About 16 min read · PDF pages 4–17 · [Short version \(Concise\)](#)

Learning goals

- Translate a vague managerial request into a stakeholder, a decision, an action horizon, a KPI, a target, a guardrail and a bounded question.
- Distinguish a KPI, a management target and a guardrail, and explain why passing one does not answer the question asked by another.
- Calculate counts, proportions and target gaps, and report a gap between two rates in percentage points rather than as a percent change.

Decision-First Analysis and the MetroMeal Brief

CONCEPT What Decision-First Analysis Means

Decision-first analysis means defining the decision before touching the data. Name the **stakeholder** (the person who will act), the **decision** (the realistic choice), the **action horizon** (when the choice must be made), the **outcome** that matters and the **evidence** needed. A **KPI**, or key performance indicator, is a quantified measure used to monitor an objective and support that decision. A KPI informs a decision; it is not the decision. Starting here prevents a polished chart that answers a question nobody asked. If the stakeholder cannot name an action the evidence could change, clarify the decision first.

CONCEPT How the Frame Works: Who, What, and How

The session builds the frame with **Who, What, and How**: who will use the communication, what they need to know or do, and how the evidence will be delivered, live or in a document read without a presenter. The frame also separates two periods. The **observed window** is when the records were collected; the **action horizon** is when the decision takes effect. Data from the first cannot by themselves describe the second. Finally, draft a **provisional claim** before choosing a chart, and mark the words that need data support and the words that would imply a cause.

WORKED EXAMPLE Worked Evidence: Framing the MetroMeal Delivery Case

MetroMeal Delivery is the session's hypothetical running case. You are the operations analyst supporting the regional operations director, and the problem is a possible decline in on-time delivery. The vague request "Show whether delivery performance is getting worse" becomes: "Should the regional operations director investigate and potentially adjust weekend staffing for the next scheduling cycle, given weekly on-time delivery performance relative to the 85% target and the 3% complaint-rate guardrail?" The data hold one row per eligible completed delivery, August 3 through August 30, 2026: 2,000 rows, 500 per week. Constraints are a fixed staffing budget, customer privacy, incomplete causal information and protecting complaint performance.

WORKED EVIDENCE **Population:** Eligible completed MetroMeal deliveries (hypothetical teaching case)

Period: August 3 through August 30, 2026

Unit: Decision-frame elements

Frame element	MetroMeal answer
Stakeholder	Regional operations director
Decision	Whether to investigate and potentially adjust weekend staffing
Action horizon	Before the next scheduling cycle
KPI	Weekly on-time delivery rate
Target	At least 85.0% on time
Guardrail	Complaint rate no greater than 3.0%
Evidence boundary	Four descriptive weeks; no evidence that isolates a cause

Interpretation: The frame names who acts, what they may do and by when, before any rate or chart is produced.

CONCEPT Interpreting the Frame

The frame shows what the analysis can and cannot do. It can show whether the weekly on-time rate met the 85.0% target and whether complaints stayed within 3.0%. It can support a decision to **investigate** weekend staffing before the next cycle. It cannot show that staffing caused any change, because the records are **observational and descriptive**: they record what happened, not why. Weather, restaurant delays, order mix or staffing could each be involved, and nothing in the case separates them. That is why the frame says "investigate" rather than "fix".

⚠️ COMMON MISTAKE Common Misconception: The Chart Is the Deliverable

A widespread error is treating the chart as the product. Given "Show whether delivery performance is getting worse," an analyst produces a dramatic bar chart. But "getting worse" compares periods, and the request names no periods, deliveries or measure; different choices give different answers from the same records. A chart built without a decision also drifts toward the most striking pattern. The session's example draft, "Performance collapsed because staffing was inadequate," pairs an undefined dramatic word with a causal claim the data cannot support. The corrected sequence is decision, measure, records, calculation, chart, claim.



LIMITS AND CONDITIONS

Limits of a Decision Frame

A frame improves focus but creates no evidence. A target is useful only when its owner, value, scope and effective period are known. The frame can also be wrong: if the director's real choice concerns weekday dispatch, a perfect weekend analysis misses the point, so confirm the decision with the stakeholder. Framing cannot fix the records either; whether the 500 weekly deliveries were recorded correctly is examined in Session 3. When data are observational and descriptive, frame the claim as what occurred in the observed window, not why it occurred.



GUIDED PRACTICE

Guided Practice: Framing the HarborCare Clinic Decision

HarborCare is a made-up clinic from Lab Part A; its numbers are not MetroMeal data. The clinic manager monitors the monthly no-show rate among eligible appointments, with a maximum of 10.0% no-shows and a complaint guardrail of 4.0%. One month shows 46 no-shows and 12 complaints among 400 eligible appointments. Task 1: Name the stakeholder, a decision limited to investigating or revising reminder scheduling, and the action horizon. Task 2: Calculate both rates and compare each with its own threshold. Task 3: Write one supported claim, one trend or causal claim to remove, and a revised question. Check: $46 \div 400 = 11.5\%$; $12 \div 400 = 3.0\%$.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.



TRANSFER

Changed-Case Transfer: OmniCourier Express

The session blog offers a hypothetical case. OmniCourier Express evaluates 800 dispatches in Week 4: 680 on time against a 90.0% target, and 12 damaged packages against a 2.0% maximum guardrail. Delayed dispatches show a median delay of 10.0 minutes and a mean of 18.5 minutes. The regional director is considering weekend staffing. Build the full frame, decide whether the evidence supports investigating or changing staffing, and rewrite a draft that says "weekend staffing caused the delays." Name one piece of evidence missing from a single week.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

CONCEPT The Rate Behind a KPI

$$\text{rate} = \frac{\text{qualifying count}}{\text{eligible count}}, \quad \text{on-time rate} = \frac{O}{E}, \quad \text{complaint rate} = \frac{C}{E}$$

SYMBOLS AND PRONUNCIATION

- O (say "oh"): number of eligible deliveries that met the promised-time rule
- C (say "see"): number of eligible deliveries with a recorded complaint
- E (say "ee"): number of eligible completed deliveries in the same week

INPUTS AND ASSUMPTIONS

Counts taken from the same eligible population and the same week. The numerator must be a subset of the denominator.

Worked steps

- 1 Step 1: Count eligible completed deliveries for the week: $E = 500$.
- 2 Step 2: Count on-time deliveries among them: $O = 392$; count complaint deliveries among them: $C = 14$.
- 3 Step 3: Divide: $392/500 = 0.784$ and $14/500 = 0.028$; display as 78.4% and 2.8%.

WORKED RESULT

Week of August 24: on-time rate 78.4%; complaint rate 2.8%.

WHAT THE RESULT MEANS

Both rates share the 500-delivery denominator; the complaint rate is not $14 \div 392$. The rate describes one week and says nothing yet about why 108 deliveries were late.

WHEN TO USE IT

When success relative to opportunity matters, or when the eligible total varies between periods.

LIMITS AND CONDITIONS

Adding ineligible records, such as canceled or test orders, enlarges the denominator and understates the on-time share.

HOW TO CHECK THE REFERENCE

Confirm $392 + 108 = 500$, then divide 392 by 500 yourself instead of copying a dashboard value.



KEY TERMS

Key terms

Stakeholder:

The person or role who will use the evidence to make a decision.

Decision-First Analysis:

An approach that defines the stakeholder, choice, timing, outcome, comparison and evidence needs before selecting a measure or chart.

Action Horizon:

The time by which a decision must be made or take effect.

Observed Window:

The historical period in which the analyzed records were collected.

Key Performance Indicator (KPI):

A quantified measure used to monitor an objective and support a decision.



STUDY CHECK

Put the learning into action

- Rewrite a vague data request as a decision question that names the stakeholder, decision, time window, outcome, comparison, target and guardrail.
- Use Who, What, and How to identify the audience, what it needs to do, and how the evidence will be delivered.
- Draft a provisional claim before building a chart, and mark each word that needs data support or that implies a cause.

Counts, Proportions, and Percentage-Point Target Gaps

CONCEPT KPI, Target, and Guardrail as Separate Roles

Every MetroMeal number has one managerial role. The **KPI** is the monitored measure: the weekly on-time delivery rate. The **target** is the management benchmark: at least 85.0% on time. The **guardrail** is a second measure that should not deteriorate while the KPI is pursued: a complaint rate no greater than 3.0%. A target is chosen by management, not calculated from the records. The session's appendix adds that the 85.0% target is a minimum **floor**, so a miss should trigger an operational review, while the 3.0% guardrail is a maximum **ceiling**, so a breach should trigger customer-impact checks.

GO DEEPER The Mechanism: Count, Proportion, and Target Gap

Three calculations turn records into KPI evidence. A **count** is a frequency, such as 392 on-time deliveries. A **proportion** divides the count by the eligible total. A **target gap** subtracts the target from the observed rate; when both are percentages, the direct difference is measured in **percentage points**. A negative gap means below target, not a negative rate. **Percent change** instead divides the difference by a baseline and describes relative size. Changing the baseline changes the percent change but not the percentage-point gap, so the session reports KPI gaps in points.

Σ KEY EQUATION Worked Calculation: The Latest MetroMeal Week

For the week starting August 24, 2026, MetroMeal recorded 500 eligible completed deliveries: 392 on time and 108 late. First reconcile: $392 + 108 = 500$. The on-time rate is $392 \div 500 = 78.4\%$, and the late rate is $108 \div 500 = 21.6\%$. The target gap is $78.4\% - 85.0\% = -6.6$ percentage points. The guardrail is checked separately: $14 \text{ complaints} \div 500 = 2.8\%$, within the 3.0% maximum.

WORKED EVIDENCE **Population:** Eligible completed MetroMeal deliveries, latest week

Period: August 24 through August 30, 2026

Unit: Deliveries and percentages

Quantity	Calculation	Result	Benchmark	Status
Denominator check	$392 + 108$	500	500 eligible	Reconciles
On-time rate	$392 \div 500$	78.4%	At least 85.0%	Below target
Target gap	$78.4 - 85.0$	-6.6 points	0 or above	Shortfall
Complaint rate	$14 \div 500$	2.8%	At most 3.0%	Within guardrail

Interpretation: The latest week missed the on-time target by 6.6 percentage points while complaints stayed inside their ceiling.

CONCEPT Interpreting a KPI Miss and a Guardrail Pass

Keep the two results separate. The KPI says 78.4% of 500 eligible deliveries in the latest week met the promised-time rule, 6.6 points below the 85.0% floor. The guardrail says complaints, at 2.8%, stayed below the 3.0% ceiling. Passing the guardrail does not cancel the KPI miss, because the measures answer different questions. Write the unit, population and week into every sentence: "78.4% of 500 eligible completed deliveries in the week of August 24 were on time." A bare "78.4%" invites readers to assume the wrong population, such as all customers.

⚠️ COMMON MISTAKE Common Misconception: Calling -6.6 a "6.6% Drop"

The most frequent error calls the gap "a 6.6% drop". The gap came from subtraction, so its unit is percentage points. Percent change divides by a baseline: $(0.784 - 0.850) \div 0.850 \times 100 = -7.76\%$, which the lab rounds to -7.8%. A second error reads the guardrail as good news about delivery: "The guardrail passed, so performance is fine." The two thresholds protect different outcomes. A third error uses the wrong complaint denominator: $14 \div 392 = 3.6\%$, which would falsely suggest a breach. Complaints are counted among all 500 eligible deliveries.

🚧 LIMITS AND CONDITIONS Limits of Counts, Proportions, and Gaps

These calculations describe one week well but have boundaries. A difference from target is not automatically statistically significant or managerially important; significance testing comes later in the course. One week shows a level, not a trend, because "worsening" compares periods. The on-time

rate also hides severity, treating a delivery one minute late like one 50 minutes late, which is why Section 2.2 adds late-minute summaries. And a guardrail can miss harms: complaints may not capture excluded addresses, pressured couriers or customers who never complain.



GUIDED PRACTICE

Guided Practice: Reading the Dashboard Numbers

Use the MetroMeal slide values from Lab Part B: 392 on time among 500 eligible, a target of 0.85, 14 complaints and a guardrail of 0.03. Task 1: Calculate the on-time rate, the target gap in percentage points and the complaint rate. Task 2: Write a bounded decision question for the director. Task 3: Write one supported claim naming the target miss and the guardrail status. Task 4: Write the staffing-cause claim the records cannot support, and name one source to check before changing staffing, such as rosters, traffic, weather, region or order mix. Check: 78.4%, -6.6 and 2.8%.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.



TRANSFER

Changed-Case Transfer: Percentage Points in a Different Setting

Apply the distinction to two session examples. First, the blog compares an observed 85.0% rate with a 90.0% target: calculate the gap in points and the relative shortfall. Second, the industry appendix reports an advertising-plan audience rising from 40 million to 70 million monthly active users: calculate the absolute increase and relative growth, and explain why "30 percentage points" is wrong. Check: -5.0 points and -5.56%; 30 million users and 75% growth.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

CONCEPT

Percentage-Point Gap and Percent Change

$$\text{gap}_{pp} = (r - T) \times 100, \quad \text{percent change} = \frac{r - T}{T} \times 100$$

SYMBOLS AND PRONUNCIATION

- r (say "ar"): observed rate, stored as a proportion such as 0.784
- T (say "tee"): target rate, stored as a proportion such as 0.85
- gap_{pp} (say "gap in percentage points"): direct difference between the two levels

INPUTS AND ASSUMPTIONS

Both rates are proportions of the same kind of event over comparable populations. The target is the baseline for percent change.

Worked steps

- 1 Step 1: Subtract the target from the observed rate: $0.784 - 0.850 = -0.066$.
- 2 Step 2: Multiply by 100 for percentage points: $-0.066 \times 100 = -6.6$.
- 3 Step 3: For percent change, divide the difference by the target: $-0.066/0.850 = -0.0776$, or -7.76% .

WORKED RESULT

Gap = -6.6 percentage points; relative shortfall = -7.76% of the target level.

WHAT THE RESULT MEANS

Excel stores 78.4% as 0.784, so the lab formula `=(B11-B8)*100` multiplies by 100 on purpose. If the cell shows -0.066 , the multiplication is missing.

WHEN TO USE IT

Percentage points for a KPI's distance from target; percent change for relative growth from a named baseline.

LIMITS AND CONDITIONS

Neither number says whether the gap is meaningful for the business or larger than normal week-to-week variation.

HOW TO CHECK THE REFERENCE

Change the baseline to 0.876 and confirm that only the percent change moves.



KEY TERMS

Key terms

Target:

A management benchmark or desired level for a KPI; it is chosen, not calculated from the records.

Guardrail:

A measure that should not deteriorate while the primary KPI is pursued, with its own limit.

Count:

The number of qualifying events, such as on-time deliveries.

Proportion:

A qualifying count divided by an eligible total, ranging from 0 to 1 and often shown as a percentage.

Percentage Point:

The unit of a direct difference between two percentages.

 STUDY CHECK**Put the learning into action**

- Calculate a KPI rate and a guardrail rate from counts that share the same eligible denominator.
- Report target gaps in percentage points and reserve percent change for relative comparisons with a named baseline.
- Check the KPI and the guardrail separately, and report both results together.

 REFERENCE SUMMARY**Reference summary**

Use these concise reminders after reading the complete explanations above.

CONCEPT **Start With the Decision****NAME THE FRAME**

- Stakeholder, decision, action horizon, outcome, comparison and evidence come before any chart.

METROMEAL'S FRAME

- The regional operations director decides whether to investigate and potentially adjust weekend staffing before the next scheduling cycle.

KEEP THE BOUNDARY

- Four descriptive weeks can support an investigation. They cannot show that staffing caused a result.

CONCEPT One Role per Number

KPI, TARGET, GUARDRAIL

- KPI: weekly on-time rate. Target: at least 85.0%. Guardrail: complaint rate at most 3.0%.

FLOOR AND CEILING

- The target is a minimum floor; the guardrail is a maximum ceiling. Higher on-time is better; lower complaints are better.

CONCEPT The Latest Week in Numbers

RECONCILE, THEN DIVIDE

- 392 on time + 108 late = 500 eligible. On-time rate: $392 \div 500 = 78.4\%$. Complaint rate: $14 \div 500 = 2.8\%$.

REPORT BOTH RESULTS

- The KPI missed its target; the guardrail stayed within its limit. One result does not cancel the other.

CONCEPT Points Versus Percent

SUBTRACT FOR POINTS

- $78.4\% - 85.0\% = -6.6$ percentage points.

DIVIDE FOR PERCENT CHANGE

- $-6.6 \div 85.0 = -7.76\%$. Name the baseline whenever you report a percent change.

STUDY CHECK Study check: Decision Frame, KPIs, Targets, and Guardrails

Use this ungraded check after reading. Close the explanation before responding, then reopen it to identify a specific correction.

1. Explain one central idea in your own words and connect it to a worked example above.
2. Reconstruct the example's inputs, sequence and interpretation; state what evidence supports each step.
3. Change one assumption or condition. Explain whether the same approach still applies.
4. Name a likely mistake and the check you would use to detect it.

My response

Write your explanation, evidence and changed-condition reasoning here.

MY READINESS

Choose one: **Can explain independently** / **Can explain with the reference**
/ Need to revisit.

Record the passage to revisit and the next action you will take.

Carry one unresolved question into the next section.

CONNECT TO THE LAB

Connect this section to the lab

[Open Part A in the lab guide →](#)

Session lab reference

S02P (ungraded practice): Can you trace a business story from decision through KPI, grain, chart, claim, and validation?

Open the assigned task

Lab Part A: What decision can the KPI inform? and Lab Part B: Is MetroMeal's latest week below target? (hand-off at slide 19, return at slide 20; 3 minutes each)

Student lab file

`S02P_Learning_Workbook.xlsx` (card guide: `S02P_Student_Lab.html`)

Workbook location

'Class_Practice'!B6:B17 (Part A) and 'Slide_Practice'!B6:B17 (Part B)

SPSS location

Block H1

Primary task

Frame the HarborCare decision and check its no-show rate against the target and the complaint guardrail.

Supporting task

Calculate MetroMeal's latest on-time rate, target gap in percentage points and complaint rate, then separate a supported claim from a causal claim.

Excel task

In Class_Practice, enter the thresholds and counts and use `=B12/B13` and `=B14/B13` (checks: 11.5% and 3.0%). In Slide_Practice, use `=B6/B7`, `=(B11-B8)*100` and `=B9/B7` (checks: 78.4%, -6.6 and 2.8%).

SPSS task

Run the supplied H1 block when directed and compare its LIST output (.784, -6.6, .028) with Slide_Practice!B11:B13.

Bring back to the discussion

One bounded decision question, one supported descriptive claim and one causal claim to remove.

Follow the lab's own prerequisites, instructions and submission rules. The Toolkit is a study reference and does not change those requirements.

My connection

Which idea from this section supports my lab decision, and what evidence will I check?

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2.2 · Reproducible KPI Definitions, Aggregation, and Center and Position

IN CLASS

Slides 20–38

Reproducible KPI Specification 21–22

Count, Proportion, and Target Gap 23–25

Percentage Points vs Percent Change 26–27

Reproduce the KPI in Excel 365 28

Aggregation from Counts 32–34

PRACTISE IT IN THE LAB

- Part C: Is the KPI calculation reproducible? · 3 min
- Part D: What is the four-week on-time rate, built from counts? · 3 min · SPSS Block H2: weekly and aggregate KPI calculations
- Part L: How did each week compare with the target and the guardrail? · 4 min · SPSS Guided Step 2: weekly KPI, guardrail, aggregate, and late-delivery checks

About 16 min read · PDF pages 18–31 · [Short version \(Concise\)](#)

Learning goals

- State the unit, eligible population, time window, grain, numerator, denominator and direction of a KPI so another analyst can reproduce it.
- Combine weekly rates by summing counts before dividing, and explain when an average of rates gives the wrong answer.
- Interpret the mean, median and 90th percentile of late minutes as descriptions of center and position, and name what they cannot establish.

Reproducible KPI Specification and Count-First Aggregation

CONCEPT What Makes a KPI Reproducible

A KPI is **reproducible** when another analyst could compute the same number from your written definition alone. A formula is not enough, because "on-time rate" can hide different populations, windows and rules. A **KPI specification** states the name, purpose, unit, eligible population, time window, grain, inclusion and exclusion rules, numerator or aggregation rule, denominator, direction

of improvement, target, guardrail and refresh schedule. The **eligible population** is the set of records allowed into the denominator. Locking these rules before calculating means a later change in the number reflects operations, not a quiet change in definition.

CONCEPT How Aggregation From Counts Works

Count-first aggregation adds the qualifying counts across periods, adds the eligible counts, and divides once, so each delivery counts once. Averaging weekly percentages instead gives each **week** equal influence regardless of volume. A **delivery-weighted** rate asks what share of all eligible deliveries arrived on time; a **week-weighted** average asks what the average week looked like. They agree only when every week has the same denominator. The session's rule: do not average weekly percentages for an overall rate unless denominators are equal or an unweighted weekly average is explicitly the intended KPI.

WORKED EXAMPLE Worked Evidence: MetroMeal's Four Weeks

The lab rebuilds the weekly table from the 2,000 rows of `orders_story.csv`. Each week has 500 eligible deliveries, with 420, 438, 425 and 392 on time. Summing counts gives 1,675 of 2,000, or 83.75%, 1.25 points below target. Complaints total 47, a four-week rate of 2.35%. Averaging the four weekly rates also gives 83.75%, but only because the denominators are equal.

WORKED EVIDENCE **Population:** 2,000 eligible completed MetroMeal deliveries in `orders_story.csv`

Period: August 3 through August 30, 2026

Unit: Deliveries per week and percentages

Week start	Eligible	On time	On-time rate	Gap (points)	Complaints	Complaint rate
2026-08-03	500	420	84.0%	-1.0	10	2.0%
2026-08-10	500	438	87.6%	+2.6	11	2.2%
2026-08-17	500	425	85.0%	0.0	12	2.4%
2026-08-24	500	392	78.4%	-6.6	14	2.8%
Four-week aggregate	2,000	1,675	83.75%	-1.25	47	2.35%

Interpretation: Three weeks at or near target and one week well below it pool to 83.75%, so the total alone hides the weak latest week.

CONCEPT Interpreting the Weekly and Pooled Results

Read the table two ways. Down the rows, the latest week is the only one clearly below target, and its complaint rate is the highest of the four, though still within 3.0%. The aggregate row, 83.75%, is correct for all 2,000 deliveries but answers a different question: three strong weeks dilute the weak one. The director therefore needs the weekly rows, not only the total. Four weekly points describe a short sequence; they do not establish a stable trend or explain why the latest week was lower.

COMMON MISTAKE Common Misconception: Averaging Rates Is Always Safe

Because MetroMeal's weeks are equal in size, =AVERAGE(...) of the weekly rates happens to give 83.75%. With unequal weeks it fails. The lab's example: 90 of 100 on time (90.0%) and 300 of 400 (75.0%) average to 82.5%, but only 390 of 500 deliveries were on time, 78.0%. A second misconception is that definitions can shift quietly. Adding abandoned calls to a resolution rate's denominator creates a new KPI version, and the session notes that if "on time" changed from 45 to 50 minutes, MetroMeal's rate could rise with no operational improvement.



LIMITS AND CONDITIONS

Limits of Specifications and Pooled Rates

A specification makes a KPI repeatable, not correct: a wrongly assigned eligibility flag is repeated faithfully. A pooled rate hides timing, so 83.75% cannot show whether the shortfall was spread out or concentrated. Weighting is a choice that must follow the question. When a definition must change, version it: record the new rule, effective date, reason and approver, recalculate earlier results where the data allow, and warn users when comparisons break.



GUIDED PRACTICE

Guided Practice: Is NorthLine's KPI Reproducible?

NorthLine is a made-up support center from Lab Part C. In September 2026, 684 of 760 eligible calls were resolved within 5 minutes; the target is at least 92.0%. Task 1: State the population, window, numerator and denominator in words. Task 2: Calculate the rate and the gap in percentage points. Task 3: Decide whether a count or a proportion answers the service question. Task 4: Write a one-line specification. Task 5: Explain how a changed eligibility rule could change the decision. Check: 90.0%; -2.0 points.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.



TRANSFER

Changed-Case Transfer: A Larger Latest Week

The lab's optional Topic 5 changes one condition: MetroMeal's latest week becomes 784 on time out of 1,000 eligible, still 78.4%, with the first three weeks unchanged. Calculate the average of the four rates and the count-first rate, explain which counts every delivery once, and describe a situation in which a week-weighted average would be the intended KPI. Check: 83.75% versus $2,067 \div 2,500 = 82.68\%$.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

CONCEPT Count-First Aggregation

$$\text{Four-week rate} = \frac{\sum_{w=1}^4 O_w}{\sum_{w=1}^4 E_w} \quad \text{not, in general,} \quad \frac{1}{4} \sum_{w=1}^4 \frac{O_w}{E_w}$$

SYMBOLS AND PRONUNCIATION

- w (say "double-u"): week number, from 1 to 4
- O_w (say "oh sub double-u"): on-time deliveries in week w
- E_w (say "ee sub double-u"): eligible deliveries in week w

INPUTS AND ASSUMPTIONS

Weekly qualifying and eligible counts built under one unchanged KPI definition.

Worked steps

- 1 Step 1: Add the on-time counts: $420 + 438 + 425 + 392 = 1,675$.
- 2 Step 2: Add the eligible counts: $500 \times 4 = 2,000$.
- 3 Step 3: Divide once: $1,675/2,000 = 0.8375$, or 83.75%.

WORKED RESULT

Four-week on-time rate = 83.75%, a gap of -1.25 percentage points against 85.0%.

WHAT THE RESULT MEANS

Each eligible delivery carries equal weight. The right-hand expression gives each week equal weight instead. The two coincide here only because every E_w equals 500. With 100 and 400 deliveries, they give 82.5% and 78.0%.

WHEN TO USE IT

Use count-first aggregation for any overall rate built from subgroups: weeks, regions or channels.

LIMITS AND CONDITIONS

The pooled rate hides differences between the subgroups. Report the subgroup rows alongside it.

HOW TO CHECK THE REFERENCE

In SPSS Block H2, AGGREGATE sums the counts and lists overall_rate .8375; Slide_Practice!B34 should show 83.75%.



KEY TERMS

Key terms

KPI Specification:

The written facts that let another analyst reproduce the same measure: purpose, unit, population, window, grain, eligibility, calculation, direction, benchmark and refresh schedule.

Eligible Population:

The records allowed into a KPI's denominator under its rules.

Count-First Aggregation:

Summing qualifying and eligible counts across subgroups before dividing once.

Delivery-Weighted Rate:

An overall rate in which each delivery carries equal influence.

Week-Weighted Average:

An average of weekly rates in which each week carries equal influence regardless of volume.



STUDY CHECK

Put the learning into action

- Write a complete KPI specification before calculating, and keep it beside the result.
- Build overall rates by summing counts first, and show the subgroup rows beside the total.
- Treat any change in eligibility, window or threshold as a new KPI version and disclose it.

Center and Position: Describing Late Deliveries



GO DEEPER

Why a Rate Needs Center and Position

The on-time rate counts how often deliveries were late, not how late they were. To describe severity, the session summarizes **late minutes** among late deliveries. The **mean** is the total divided by the count and uses every value. The **median** is the ordered middle value, or the average of the two middle values

when the count is even. A **percentile** is a position statement: the 90th percentile is a threshold at or below which about 90% of values fall, under a stated convention. Mean and median describe **center**; the percentile describes **position**.

CONCEPT How the Three Summaries Respond to Long Delays

The mean adds every value, so a few long delays pull it upward. The median depends on the middle positions and moves much less. The 90th percentile sits in the upper part of the ordered values and shows how long the longer delays are. A mean above the median can be consistent with a **right-skewed** distribution, one with a longer tail of high values, but the session treats this only as a prompt to inspect the distribution. Percentiles depend on a convention: the lab uses Excel's `PERCENTILE.INC`, and SPSS's rule can differ slightly.

WORKED EXAMPLE Worked Evidence: Late Minutes in the Latest Week

In the week of August 24, 108 of 500 eligible deliveries were late. The session reports a mean of 12.4 minutes, a median of 7 and a 90th percentile of 31. To make these traceable, this toolkit tabulated `late_minutes` for those 108 rows of `orders_story.csv` (week_start 2026-08-24, eligible, late_minutes above 0). Every promised time is 45 minutes, and `late_minutes` is delivered time minus 45. The total is 1,339.2 minutes, and $1,339.2 \div 108 = 12.4$.

WORKED EVIDENCE **Population:** 108 late eligible deliveries in the latest week (computed from `orders_story.csv`)

Period: August 24 through August 30, 2026

Unit: Minutes late per delivery

Minutes late	Deliveries	Sorted positions	Minutes contributed
2.0	20	1 to 20	40.0
7.0	35	21 to 55	245.0
12.0	41	56 to 96	492.0
31.0	2	97 to 98	62.0
50.0	9	99 to 107	450.0
50.2	1	108	50.2

Interpretation: Most late deliveries were 2 to 12 minutes late, but ten were about 50 minutes late, which lifts the mean well above the median.

CONCEPT Interpreting the Mean, Median, and 90th Percentile

With 108 values, the median averages the 54th and 55th sorted values; both are 7, so half the late deliveries were 7 minutes late or less. The mean, 12.4, is 5.4 minutes higher because ten deliveries were about 50 minutes late. The 90th percentile, 31 minutes, places the upper end of severity near half an hour; repeated values mean the split need not be exactly 90/10. The median describes the typical late experience, the mean the average burden, and the percentile the longer delays. Together they justify inspecting the full distribution in Session 3, not a conclusion about causes.

COMMON MISTAKE Common Misconception: One Summary Tells the Whole Story

A frequent error reports the mean alone as the "typical" delay. Here 96 of the 108 late deliveries were 12 minutes late or less, below the 12.4 mean. The opposite error reports only the reassuring median and hides the ten deliveries near 50 minutes. A third error forgets the subset: these summaries describe the 108 **late** deliveries in one week, not all 500; across all eligible deliveries the mean late time would be far smaller because 392 contribute zero. State the population, window and subset for every summary.



LIMITS AND CONDITIONS

Limits of Center and Position Summaries

Three numbers cannot show whether long delays clustered on certain days, regions or weekends, and they cannot identify a cause. Percentiles vary by software convention, especially with repeated values and small groups. The session reserves claims about skew, outliers, spread and shape for Session 3. Choose the form by the decision: a mean when every value's contribution matters, a median or percentile when skew, service thresholds or tail experience matter.



GUIDED PRACTICE

Guided Practice: Choosing the Decision-Useful Statement

The session asks which statement is most useful to the director: 392 on-time deliveries; a 78.4% on-time rate; a 7-minute median delay among late deliveries; or a 31-minute 90th percentile delay. Choose one for (1) a staffing decision about volume, (2) a service promise to customers and (3) a review of the worst delays. For each choice, write the population, window and unit it describes, using the table in 2.2.2.3.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.



TRANSFER

Changed-Case Transfer: Earlier MetroMeal Weeks

Apply the same summaries to earlier weeks, as in the lab's optional Topic 4. This toolkit computed them from `orders_story.csv` with pandas (late_minutes above 0 within each week; 90th percentile by the PERCENTILE.INC rule). August 3: 80 late, mean 11.4, median 11, 90th percentile 19. August 10: 62 late, mean 11.6, median 12, 90th percentile 19. August 17: 75 late, mean 11.4, median 11, 90th percentile 18.6. August 24: 108 late, mean 12.4, median 7, 90th percentile 31. In which week is the mean furthest above the median, what would you inspect next, and what can you not yet conclude?

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

CONCEPT Mean, Median, and Inclusive Percentile

$$x = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{x} = \frac{x_{(n/2)} + x_{(n/2+1)}}{2} \text{ for even } n, \quad \text{position}_p = 1 + p(n-1)$$

SYMBOLS AND PRONUNCIATION

- $\bar{x}$ (say "x-bar"): mean late minutes
- $\tilde{x}$ (say "x-tilde"): median late minutes
- n (say "en"): number of late deliveries, here 108
- $x_{(k)}$: the value at sorted position k
- p : the percentile as a proportion, here 0.9

INPUTS AND ASSUMPTIONS

Late minutes for eligible late deliveries in one week, sorted from smallest to largest.

Worked steps

- 1 Step 1: Mean: $1,339.2/108 = 12.4$ minutes.
- 2 Step 2: Median: $n = 108$ is even, so average positions 54 and 55: $(7 + 7)/2 = 7$ minutes.
- 3 Step 3: 90th percentile position: $1 + 0.9 \times 107 = 97.3$, between positions 97 and 98.
- 4 Step 4: Both positions hold 31, so the 90th percentile is 31 minutes.

WORKED RESULT

Mean 12.4 minutes; median 7 minutes; 90th percentile 31 minutes.

WHAT THE RESULT MEANS

The position rule shown is the inclusive rule used by Excel's `PERCENTILE.INC`. Other rules interpolate differently and can give slightly different cut points. The median needs only the middle positions; the mean needs every value.

WHEN TO USE IT

When a KPI must describe how severe a problem was, not just how often it occurred.

LIMITS AND CONDITIONS

These summaries describe the observed late deliveries only. They do not establish skew, outliers or causes without inspecting the full distribution.

HOW TO CHECK THE REFERENCE

KPI_Build!D18:F18 should show 12.4, 7 and 31. In SPSS, Guided Step 2 runs EXAMINE on the same 108 deliveries; its percentile rule may differ slightly from 31.



KEY TERMS

Key terms

Mean:

The sum of eligible numerical values divided by their count; every magnitude affects it.

Median:

The ordered middle value, or the average of the two middle values when the count is even.

Percentile:

A threshold describing position in an ordered distribution under a stated convention.

Right-Skewed:

Describes a distribution with a longer tail of high values; a mean above the median is consistent with it but does not prove it.

Late Minutes:

Delivered time minus promised time for a late delivery; zero for an on-time delivery in this file.



STUDY CHECK

Put the learning into action

- Pair a frequency KPI such as the on-time rate with a severity summary such as the median or 90th percentile of late minutes.
- State the population, window, subset and percentile convention beside every center or position summary.
- Treat a mean well above the median as a reason to inspect the distribution, not as a finding.



REFERENCE SUMMARY

Reference summary

Use these concise reminders after reading the complete explanations above.

CONCEPT Write the Specification

EVERYTHING ANOTHER ANALYST NEEDS

- Name, purpose, unit, population, window, grain, eligibility, numerator, denominator, direction, target, guardrail and refresh schedule.

CHANGES CREATE VERSIONS

- A new eligibility rule or threshold is a new KPI version. Disclose it before comparing results.

CONCEPT Sum Counts, Then Divide

METROMEAL'S FOUR WEEKS

- $1,675 \div 2,000 = 83.75\%$, -1.25 points against target.

WHY NOT AVERAGE THE RATES

- 90 of 100 and 300 of 400 average to 82.5%, but the deliveries give $390 \div 500 = 78.0\%$.

CONCEPT Center and Position

THREE SUMMARIES OF 108 LATE DELIVERIES

- Mean 12.4 minutes; median 7 minutes; 90th percentile 31 minutes (PERCENTILE.INC).

WHAT THE GAP SUGGESTS

- The mean is 5.4 minutes above the median because ten deliveries were about 50 minutes late. Inspect the distribution before claiming skew.

CONCEPT Name the Subset

SAY WHAT EACH NUMBER DESCRIBES

- The late-minute summaries describe late deliveries only, in one week. The on-time rate describes all eligible deliveries in that week.

 STUDY CHECK

Study check: Reproducible KPI Definitions, Aggregation, and Center and Position

Use this ungraded check after reading. Close the explanation before responding, then reopen it to identify a specific correction.

1. Explain one central idea in your own words and connect it to a worked example above.
2. Reconstruct the example's inputs, sequence and interpretation; state what evidence supports each step.
3. Change one assumption or condition. Explain whether the same approach still applies.
4. Name a likely mistake and the check you would use to detect it.

My response

Write your explanation, evidence and changed-condition reasoning here.

MY READINESS

Choose one: **Can explain independently** / **Can explain with the reference**
/ Need to revisit.

Record the passage to revisit and the next action you will take.

Carry one unresolved question into the next section.

 CONNECT TO THE LAB

Connect this section to the lab

Open Part C in the lab guide →

Session lab reference

S02P (ungraded practice): Can you trace a business story from decision through KPI, grain, chart, claim, and validation?

Open the assigned task

Lab Part C: Is the KPI calculation reproducible? and Lab Part D: What is the four-week on-time rate, built from counts? (hand-off at slide 38, return at slide 39), then Lab Part L: How did each week compare with the target and the guardrail? (guided build, 4 minutes)

Student lab file

S02P_Learning_Workbook.xlsx (card guide: S02P_Student_Lab.html)

Workbook location

'Class_Practice'!B21:B35, 'Slide_Practice'!B21:B35 and 'KPI_Build'!B6:H18

SPSS location

Block H2 and Guided Step 2

Primary task

Build the weekly KPI table from the raw rows and the count-first four-week aggregate.

Supporting task

Write a reproducible KPI specification and summarize late minutes for the latest week.

Excel task

In KPI_Build, use COUNTIFS and SUMIFS for rows 7 to 10, =D11/C11 for the aggregate, and AVERAGEIFS, MEDIAN(FILTER(...)) and PERCENTILE.INC(FILTER(...)) in row 18. Checks: row 10 shows 500, 392, 78.4%, -6.6, 14 and 2.8%; E11 shows 83.75%; G18 shows 0.

SPSS task

Run Block H2 (overall_rate .8375) during the hand-off, and Guided Step 2 during the guided build; compare its weekly LIST and EXAMINE output with KPI_Build.

Bring back to the discussion

A checked KPI formula with units, visible numerator and denominator, target gap and one limitation.

Follow the lab's own prerequisites, instructions and submission rules. The Toolkit is a study reference and does not change those requirements.

My connection

Which idea from this section supports my lab decision, and what evidence will I check?

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2.3 · Variables, Measurement Scales, Observations, and Grain

IN CLASS

[Slides 39–54](#)

Variables and Observations 40–41

Data Type and Measurement Scale 42–43

Row Grain and Supported Claims 45–46

Mixed-Grain Join Risk 47

Visualizing the Join Fan Trap 48–49

PRACTISE IT IN THE LAB

- [Part E: What does one row represent?](#) · 3 min
- [Part F: How many orders hide in four joined rows?](#) · 3 min · SPSS Block H3: item grain, repeated order IDs, and restoration to order grain
- [Part K: What does one row of the MetroMeal file represent?](#) · 4 min · SPSS Guided Step 1: source, candidate key, grain, and eligibility

About 15 min read · PDF pages 32–44 · [Short version \(Concise\)](#)

Learning goals

- Distinguish variables from observations and classify each field separately by data type and measurement scale.
- Identify the grain of raw and aggregated tables, test a candidate key, and explain how changing grain changes the supported claim.
- Detect the mixed-grain error created by a one-to-many join and restore the correct grain before calculating a KPI.

Variables, Observations, Data Types, and Measurement Scales

CONCEPT Variables and Observations

A **variable** is a characteristic recorded for members or events, such as region or delivery minutes; it appears as a column. An **observation** is the set of values for one member or event; it appears as a row. **Cross-sectional data** describe many entities at one period; **time-series data** track a measure through

ordered periods. MetroMeal's weekly table is a four-point time series, too short to establish a stable trend. A dataset can contain every eligible event in its window and still not represent future periods, excluded events or a broader customer population.

CONCEPT Data Type Versus Measurement Scale

Each field receives two classifications. **Data type:** **categorical** (labels, codes, flags) or **numerical**, which is **discrete** when counted and **continuous** when measured. **Measurement scale:** **nominal** (names, no order), **ordinal** (ordered, unknown spacing), **interval** (equal gaps, no true zero) or **ratio** (equal gaps and a true zero, so ratios make sense). Storage format is a third, unrelated property: digits stored as numbers do not make a field numerical. The session's analogy: jersey number 10 is not twice jersey number 5.

WORKED EXAMPLE Worked Evidence: One MetroMeal Observation

Order 10482 appears in the slides and in `orders_story.csv`. Reading across its row gives one observation: customer 5241, completed Saturday, August 8, 2026 (week starting August 3), North region, 3 items, promised 45 minutes, delivered in 54, 9 minutes late, satisfaction category 2, eligible, not on time, no complaint. Completion date is a date-time variable: ordered and usable for interval calculations under defined conventions, but not an ordinary quantity just because Excel stores it as a number.

WORKED EVIDENCE **Population:** Fields in the MetroMeal delivery file (`orders_story.csv`)

Period: August 3 through August 30, 2026

Unit: One field per row of this table

Field	Order 10482 value	Data type	Measurement scale	Analytical role
order_id	10482	Categorical	Nominal	Identifier (candidate key)
region	North	Categorical	Nominal	Category
satisfaction_category	2	Categorical	Ordinal	Ordered category
item_count	3	Numerical, discrete	Ratio	Measure
delivery_minutes	54.0	Numerical, continuous	Ratio	Measure
on_time_kpi_flag	0	Categorical (binary)	Nominal	Flag

Interpretation: Digits appear in order_id, satisfaction_category and item_count, but only item_count and delivery_minutes support ordinary arithmetic.

CONCEPT Interpreting the Classifications

Classification decides which summaries are meaningful. An average of order_id values means nothing. For satisfaction_category, 2 is lower than 4, but the gap from 1 to 2 need not equal the gap from 4 to 5, so use counts, category percentages and the median. In the file the categories run 1: 47, 2: 278, 3: 557, 4: 560 and 5: 558 deliveries (counted from `orders_story.csv`). For item_count and delivery_minutes, a true zero exists, so sums, means and ratios work. A 0/1 flag is categorical, yet summing it counts the 1s and averaging it gives their proportion.

⚠ COMMON MISTAKE Common Misconception: If Excel Can Average It, It Is a Number

Excel will average any column of digits, but that does not make the result meaningful: averaged order IDs, postal codes or ordinal codes have no interpretation. A related error mixes the classifications: "nominal" is a scale, "discrete" is a type, so item_count is numerical discrete and ratio-scaled. A third error treats ordinal ratings as equally spaced by default. A mean rating needs an explicitly disclosed

equal-interval convention plus the category frequencies, because a mean alone can hide a split between very satisfied and very dissatisfied customers. Store identifiers as Text so formulas do not treat labels as quantities.

LIMITS AND CONDITIONS **Limits of Classification**

Some classifications belong to the construct; others depend on the analytical use. Recorded precision does not settle type: minutes recorded as whole numbers still measure continuous time. Dates need explicit conventions before subtraction. Classification tells you which operations are admissible, not whether values are accurate or complete; that is Session 3's work.

GUIDED PRACTICE **Guided Practice: Classify the BrightCart Fields**

BrightCart is a made-up retailer from Lab Part E. Record storage, data type, measurement scale and analytical role for `order_id`, `item_count`, `satisfaction_category`, `return_flag` and `region`. Then decide what you would report first for satisfaction stored as 1 to 5, and what convention you would disclose if a manager insists on a mean. Lab hints: `order_id` is digits but categorical and nominal; satisfaction codes have order but no fixed spacing, so they are ordinal.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

TRANSFER **Changed-Case Transfer: Deliveries Are Not Customers**

In `orders_story.csv`, the 2,000 deliveries come from 1,000 distinct `customer_id` values, each appearing twice (counted with pandas). The session warns that a 78.4% delivery rate does not mean 78.4% of customers received on-time service. Explain why the unit of analysis matters, define the numerator and denominator of a customer-level measure such as "share of customers whose deliveries were all on time", and explain why `customer_id` must never be averaged.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

CONCEPT Summing and Averaging a 0/1 Flag

$$E = \sum_{i=1}^N f_i, \quad \frac{1}{E} \sum_{i \in \text{eligible}} g_i = \frac{O}{E}$$

SYMBOLS AND PRONUNCIATION

- N (say "en"): rows in the Orders table, here 2,000
- f_i (say "f sub i"): eligible_kpi_flag for row i , either 0 or 1
- g_i (say "g sub i"): on_time_kpi_flag for row i , either 0 or 1
- E and O : eligible and on-time counts

INPUTS AND ASSUMPTIONS

Flags coded 1 when the condition is true and 0 otherwise, with no blanks.

Worked steps

- 1 Step 1: Sum eligible_kpi_flag over all rows: $E = 2,000$ (KPI_Build!B5).
- 2 Step 2: Within the latest week, sum on_time_kpi_flag among eligible rows: $O = 392$.
- 3 Step 3: Divide by the week's eligible count: $392/500 = 0.784$.

WORKED RESULT

The sum of a flag is a count; the average of a flag over eligible rows is the proportion of 1s.

WHAT THE RESULT MEANS

This is why a nominal 0/1 field can still feed a KPI. The arithmetic is on the coded condition, not on a measured quantity.

WHEN TO USE IT

Counting eligible, on-time or complaint deliveries with SUM, SUMIFS or a PivotTable Average.

LIMITS AND CONDITIONS

A blank or a code other than 0 and 1 breaks the logic; check the flag's values first.

HOW TO CHECK THE REFERENCE

In SPSS Guided Step 1, FREQUENCIES on eligible_kpi_flag should show 2,000 cases coded 1.



KEY TERMS

Key terms

Variable:

A recorded characteristic, represented by a column.

Observation:

One complete row of values for one member or event at the stated grain.

Data Type:

Whether values are categorical or numerical, and whether numerical values are discrete or continuous.

Measurement Scale:

Whether a field is nominal, ordinal, interval or ratio in meaning.

Identifier:

A code that names an entity or event; arithmetic on it has no useful meaning.



STUDY CHECK

Put the learning into action

- Classify every field twice, by data type and by measurement scale, in separate columns.
- Check whether arithmetic is meaningful before averaging any field that software stores as a number.
- Summarize ordinal ratings with frequencies and the median, and disclose any equal-interval convention.

Row Grain, Candidate Keys, and Mixed-Grain Join Risk

CONCEPT

What Grain Means

Grain is the meaning of one row: its entity or event, level of detail and time unit. The grain test is to complete "Each row represents one ..."; more than one incompatible ending signals mixed grain. A **candidate key** is a field, or combination of fields, that is unique at the stated grain. MetroMeal's raw

table has one row per eligible completed delivery, keyed by order_id; its weekly summary has one row per week. Delivery rows support checks on individual orders and exceptions; weekly rows support comparisons across weeks.

CONCEPT How a One-to-Many Join Changes Weight

In a **one-to-many join**, one parent row matches several child rows. Joining orders (one row per order) to items (one row per item) yields one row per item, repeating each order once per item. The session calls this the **join fan trap**: parents receive unequal weight. Repeated order IDs after the join are expected at item grain, not corrupt data; the error is computing an order-level measure on the joined rows. The repair is to aggregate the child table to one row per order first, then verify key, relationship and row count before using COUNT or AVERAGE.

WORKED EXAMPLE Worked Evidence: Two Orders, Four Rows

The session's illustration uses Order A, late with three items, and Order B, on time with one item. At order grain the on-time rate is $1 \div 2 = 50.0\%$. After joining the item table, A appears three times and B once, so a row-based rate reports $1 \div 4 = 25.0\%$. Nothing about delivery changed; the late order simply received triple weight. SPSS Block H3 uses the same structure and aggregates the four item rows back to two orders.

WORKED EVIDENCE **Population:** Two-order teaching illustration (not the MetroMeal file)

Period: Not applicable; structural example

Unit: Item rows after the join

Joined row	order_id	item_id	Order outcome	Counted as
1	A	I1	Late	Late row
2	A	I2	Late	Late row
3	A	I3	Late	Late row
4	B	I4	On time	On-time row

Interpretation: Four joined rows contain only two distinct orders, so the row-based rate of 25.0% misstates the order-level rate of 50.0%.

CONCEPT Interpreting Grain in the MetroMeal File

The MetroMeal file passes its grain test: `ROWS(Orders[order_id])` and `ROWS(UNIQUE(Orders[order_id]))` both return 2,000, so `order_id` is unique at delivery grain, and `SUM(Orders[eligible_kpi_flag])` returns 2,000. The file also shows why a join would matter: 1,675 deliveries have one item and 325 have three (from `item_count`), so a direct join to an item table would create $1,675 + 3 \times 325 = 2,650$ rows and count multi-item deliveries three times. Aggregation runs the other way: 500 delivery rows become one weekly row that cannot be taken apart again.

COMMON MISTAKE Common Misconception: Deleting Repeats Cleans the Table

Seeing repeated `order_id` values after a join, analysts often delete the "duplicates". At item grain the repeats are correct; deleting rows loses information and may keep the wrong outcome. Decide which grain the question needs, then aggregate, for example with `MAX` of an order-level flag and a count of item rows, as Block H3 does. The opposite error treats four joined rows as four orders, quietly giving customers who buy more items more say. In a table that should be at order grain, duplicate keys need investigation before deletion: they may be errors or evidence that the assumed grain is wrong.

LIMITS AND CONDITIONS Limits of the Grain Test

Matching row and distinct-key counts shows the key is unique, not that every row is real and eligible or that nothing is missing. A correctly stated grain may still not suit the question: a line-item defect decision needs item grain with a redefined population and denominator. Aggregated tables cannot support claims about individual deliveries. Use raw grain to validate records and weekly grain for weekly decisions.

GUIDED PRACTICE Guided Practice: How Many Orders Hide in Four Joined Rows?

Use Lab Part F. Enter order IDs A, A, A, B in `Slide_Practice!B47:B50` and item IDs I1 to I4 in `C47:C50`.
Task 1: Count joined rows with `=COUNTA(B47:B50)` and distinct orders with `=ROWS(UNIQUE(B47:B50))`.
Task 2: Write grain sentences for the raw file (2,000 rows) and the weekly summary (4 rows). Task 3:

State what must happen before an order-level rate is calculated. Check: 4 rows, 2 orders; Block H3 aggregates to 2 rows.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

 TRANSFER **Changed-Case Transfer: BrightCart's Order-Level Return Rate**

BrightCart has 120 order rows and 310 item rows, and a manager wants the order-level return rate. Predict the rows after joining every item to its order, state each table's grain, choose the safe grain for the KPI, and explain what goes wrong on the joined rows. Whose experience gets extra weight? Check: 310 joined rows, one per item; the KPI needs one row per order.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

CONCEPT **Key Test and Joined-Row Count**

$$N_{\text{rows}} = N_{\text{distinct}} \Rightarrow \text{key is unique at this grain}, \quad n_{\text{joined}} = \sum_{j=1}^m k_j$$

SYMBOLS AND PRONUNCIATION

- N_{rows} : number of rows in the table
- N_{distinct} : number of distinct values of the candidate key
- m : number of parent orders; k_j : number of child item rows for order j

INPUTS AND ASSUMPTIONS

A candidate key column, and for joins, the number of matching child rows per parent.

Worked steps

- 1 Step 1: MetroMeal: $N_{\text{rows}} = 2,000$ and $N_{\text{distinct}} = 2,000$, so order_id is unique at delivery grain.
- 2 Step 2: Two-order illustration: $k_A = 3$ and $k_B = 1$, so $n_{\text{joined}} = 4$.
- 3 Step 3: Row-based on-time rate after the join: $1/4 = 25.0\%$; order-level rate: $1/2 = 50.0\%$.

WORKED RESULT

The key test passes for MetroMeal; the join multiplies Order A's weight by three.

WHAT THE RESULT MEANS

Equal counts mean no key value repeats. After a one-to-many join, the row count equals the number of child rows, so row-based rates weight parents by their number of children.

WHEN TO USE IT

Before any COUNT, AVERAGE or rate on a table that came from a join.

LIMITS AND CONDITIONS

A passing key test does not prove completeness or correct eligibility.

HOW TO CHECK THE REFERENCE

KPI_Build!B3 and B4 both show 2000. In SPSS Guided Step 1, AGGREGATE by order_id produces rows_per_order with minimum and maximum 1.



KEY TERMS

Key terms

Grain:

What one row represents: its entity or event, level of detail and time unit.

Candidate Key:

A field or combination of fields that is unique at the stated grain.

One-to-Many Join:

A join where one row in one table matches several rows in another.

Join Fan Trap:

The reweighting that occurs when parent rows are repeated for each matching child row.

Mixed-Grain Risk:

The distortion that occurs when repeated entities change their analytical weight in a summary.



STUDY CHECK

Put the learning into action

- Complete the grain sentence for every table before calculating anything.

- Test the candidate key by comparing row and distinct-key counts.
- Aggregate child tables to the grain the KPI needs before joining or calculating a rate.

REFERENCE SUMMARY **Reference summary**

Use these concise reminders after reading the complete explanations above.

CONCEPT **Columns and Rows**

VARIABLES AND OBSERVATIONS

- A variable is a column; an observation is one complete row. Order 10482 is one delivery observation with several variables.

CONCEPT **Two Classifications per Field**

TYPE AND SCALE

- order_id: categorical, nominal. satisfaction_category: categorical, ordinal. item_count: numerical discrete, ratio. delivery_minutes: numerical continuous, ratio.

MEANING, NOT STORAGE

- Digits can label, rank or measure. Only measurements support ordinary arithmetic.

CONCEPT **Grain and Keys**

COMPLETE THE SENTENCE

- Raw file: each row represents one eligible completed delivery (2,000 rows). Weekly table: each row represents one week (4 rows).

KEY TEST

- 2,000 rows and 2,000 distinct order_id values: the key is unique at delivery grain.

COMMON MISTAKE **The Join Fan Trap**

SAME OUTCOMES, DIFFERENT RATE

- Order A late with 3 items, Order B on time with 1 item: 50.0% by order, 25.0% by joined row.

REPAIR

- Aggregate items to one row per order first, then calculate.

✓ STUDY CHECK

Study check: Variables, Measurement Scales, Observations, and Grain

Use this ungraded check after reading. Close the explanation before responding, then reopen it to identify a specific correction.

1. Explain one central idea in your own words and connect it to a worked example above.
2. Reconstruct the example's inputs, sequence and interpretation; state what evidence supports each step.
3. Change one assumption or condition. Explain whether the same approach still applies.
4. Name a likely mistake and the check you would use to detect it.

My response

Write your explanation, evidence and changed-condition reasoning here.

MY READINESS

Choose one: **Can explain independently** / **Can explain with the reference**
/ Need to revisit.

Record the passage to revisit and the next action you will take.

Carry one unresolved question into the next section.

✍ CONNECT TO THE LAB

Connect this section to the lab

[Open Part E in the lab guide →](#)

Session lab reference

S02P (ungraded practice): Can you trace a business story from decision through KPI, grain, chart, claim, and validation?

Open the assigned task

Lab Part E: What does one row represent? and Lab Part F: How many orders hide in four joined rows? (hand-off at slide 54, return at slide 55), then Lab Part K: What does one row of the MetroMeal file represent? (guided build)

Student lab file

S02P_Learning_Workbook.xlsx (card guide: S02P_Student_Lab.html)

Workbook location

'Class_Practice'!B39:F52, 'Slide_Practice'!B39:F52 and 'KPI_Build'!B3:B5, with the decision frame in 'Guided_Practice'!B5

SPSS location

Block H3 and Guided Step 1

Primary task

Confirm MetroMeal's key, grain and eligibility before any calculation.

Supporting task

Classify fields by type, scale and role, and diagnose the order-to-item join.

Excel task

Enter =ROWS(Orders[order_id]), =ROWS(UNIQUE(Orders[order_id])) and =SUM(Orders[eligible_kpi_flag]) in KPI_Build!B3:B5 (each shows 2000). In Class_Practice, predict joined rows with =B46 (310); in Slide_Practice, use COUNTA and ROWS(UNIQUE(...)) on B47:B50 (4 and 2).

SPSS task

Run Block H3 during the hand-off and Guided Step 1 during the guided build; read the FREQUENCIES and rows_per_order output.

Bring back to the discussion

A completed type-and-scale classification, two grain statements and one mixed-grain warning.

Follow the lab's own prerequisites, instructions and submission rules. The Toolkit is a study reference and does not change those requirements.

My connection

Which idea from this section supports my lab decision, and what evidence will I check?

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2.4 · Honest Charts, Bounded Stories, and Validation

IN CLASS

Slides 55–70

Match the Display to the Question 56–58

Axes, Baselines, and Statistical Meaning 59–61

Chart Critique and Accessibility Protocol 62–63

Apply the Critique in Excel 365 64

Bounded Chart Interpretation 65–66

PRACTISE IT IN THE LAB

- Part G: Does the chart communicate honestly? · 3 min
- Part H: What makes the MetroMeal dashboard chart misleading? · 3 min · SPSS Block H4: four weekly rates and a simple zero-baseline bar chart
- Part I: Is the data story traceable and bounded? · 3 min
- Part J: Can every number in the MetroMeal story be traced? · 3 min · SPSS Block H5: reconcile the latest evidence and calculate story values
- Part M: Does the chart show the checked numbers honestly? · 4 min · SPSS Guided Step 3: weekly on-time chart from checked values
- Part N: What should the director do next? · 3 min · SPSS Guided Step 4: values to validate against the chart title and bounded story

About 16 min read · PDF pages 45–58 · [Short version \(Concise\)](#)

Learning goals

- Select a table, bar chart or line chart according to the viewer's question, and critique a chart for baseline, denominator, color, clutter and title problems.
- Repair a misleading chart so that it shows checked values with a zero baseline, direct labels, denominators, a target line and a non-color cue.
- Write a three-sentence bounded story and validate every number in it through an independent route.

Chart Selection, Baselines, and Accessible Critique

CONCEPT Matching the Display to the Question

Choose the display by the viewer's task. A **table** supports exact lookup, a **bar chart** compares magnitudes across categories or periods, and a **line chart** emphasizes ordered progression. Horizontal bars suit long labels; order categories naturally or by a disclosed rule; never connect unordered categories with lines. Match the statistic too: a weekly rate is a proportion with a denominator, not a count or a causal effect, and a target line is a benchmark, not a fitted average. Show counts and rates together when a changing denominator could explain a rate movement.

CONCEPT How Baselines, Clutter, and Accessibility Shape Perception

Viewers judge bars by length from a common baseline, so quantitative bar charts should begin at zero; a **truncated axis** makes modest differences look large. Line charts encode position and may use a narrower range when the range, labels and context are explicit. **Clutter** is decoration, unneeded gridlines, redundant labels, 3D effects and excess color. **Emphasis** should direct attention to a decision-relevant feature without hiding context. **Accessible** charts use readable text, contrast, direct labels and cues beyond color, with a text description and data table.

WORKED EXAMPLE Worked Evidence: Critiquing and Repairing Chart A

The session's Chart A shows MetroMeal's four weekly rates as 3D bars on a 75% to 90% axis, uses red and green as the only status cues, omits denominators and is titled "Delivery performance collapses." The critique protocol checks meaning first (decision, statistic, grain, period, eligibility, numerator, denominator, source), then encoding and accessibility, then whether the title matches the data. The repair uses a zero baseline, direct labels, a labeled 85% target, one accessible emphasis on the latest week and visible denominators.

WORKED EVIDENCE **Population:** MetroMeal weekly on-time rates, 500 eligible deliveries per week

Period: August 3 through August 30, 2026

Unit: Chart features

Feature	Chart A (flawed)	Repaired chart
Axis	75.0% to 90.0%	0.0% to 100.0%
Form	3D vertical bars	2D bars with an 85.0% target line
Denominators	Omitted	Label "78.4% (n = 500)" on each bar
Status cue	Red and green only	Pattern fill or thick outline on the latest bar
Title	"Delivery performance collapses."	"Week of August 24 on-time delivery was 78.4%, 6.6 percentage points below target."

Interpretation: The data values are identical in both charts; only the encoding and wording change what a reader believes.

CONCEPT Interpreting the Repaired Chart

On the truncated axis, the 78.4% bar occupies only 3.4 points of a 15-point range and looks close to collapse. On a 0% to 100% axis it sits 9.2 points below the 87.6% week and 6.6 points below target: concerning, but far from zero. The repaired chart supports a bounded reading: the latest observed week was lower than each prior week and below target, with each rate resting on 500 deliveries. It does not show why, establish a trend from four points or forecast next week.

⚠️ COMMON MISTAKE Common Misconception: Cosmetic Cleanup Is a Repair

Fixing the colors while keeping the 75.0% baseline is not a repair; the bars still exaggerate the dip. Typing rates into the chart table instead of linking them is another error: a later correction leaves the chart disagreeing with the analysis. Reading counts as rates is a third: in the lab's RiverTown case, 54 of 60 is the same 90.0% as 72 of 80, although 54 looks smaller. Finally, zero is not required for every chart; a clearly labeled line chart may use a narrower range.

🚧 LIMITS AND CONDITIONS Limits of Chart Repair

An honest chart cannot repair a wrong eligibility rule, grain or denominator; it only displays the error clearly. Emphasis always carries a message, so ask whether highlighting the latest week is warranted or merely dramatic. If a repair changes the apparent story, revise the claim rather than keeping

dramatic wording. A chart read without a presenter needs its own title, labels, annotation, source and limitation.



GUIDED PRACTICE

Guided Practice: Is RiverTown's Chart Honest?

RiverTown is a made-up program from Lab Part G: attendance was 72 of 80 eligible residents in January, 54 of 60 in February and 30 of 50 in March, against an 85.0% target. The flawed chart starts at 50.0% and uses red and green alone. Task 1: Calculate the three rates. Task 2: Name the scale or denominator problem and the color-only problem. Task 3: List the repairs. Task 4: Write a title stating the observed March comparison without "will". Check: 90.0%, 90.0%, 60.0%.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.



TRANSFER

Changed-Case Transfer: A Line Chart for the Same Weeks

Suppose the director's question is about sequence rather than magnitude. Design a line chart of MetroMeal's four weekly rates. Decide whether a nonzero range is acceptable and which labels, target line and window disclosure are required. Explain what the line emphasizes that bars do not, and write one sentence on why four connected points do not establish a stable trend.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

CONCEPT

How a Truncated Axis Distorts Bar Length

$$\text{visible share} = \frac{v - a_{\min}}{a_{\max} - a_{\min}}, \quad \text{apparent ratio} = \frac{v_1 - a_{\min}}{v_2 - a_{\min}}$$

SYMBOLS AND PRONUNCIATION

- v : plotted value, such as 0.784

- $a_{\min}$ and $a_{\max}$ (say "a-min" and "a-max"): the lowest and highest values on the axis
- v_1 and v_2 : two plotted values being compared

INPUTS AND ASSUMPTIONS

A bar chart whose bar length starts at the axis minimum.

Worked steps

- 1 Step 1: Truncated axis, 75% to 90%: the 78.4% bar fills $(78.4 - 75)/15 = 22.7\%$ of the plot height; the 87.6% bar fills $(87.6 - 75)/15 = 84.0\%$.
- 2 Step 2: Apparent ratio of the two bars: $12.6/3.4 = 3.7$, so one bar looks almost four times the other.
- 3 Step 3: Zero baseline: $87.6/78.4 = 1.12$, so the bars differ by about 12%, matching the data.

WORKED RESULT

Truncation turns a 9.2-point difference into bars that look 3.7 times different in length.

WHAT THE RESULT MEANS

Above a zero minimum, bar length no longer represents the value, so the eye reads a false ratio.

WHEN TO USE IT

To explain why a dashboard axis must be reset before a decision.

LIMITS AND CONDITIONS

The calculation applies to length encodings such as bars. Line charts encode position and can use narrower ranges with clear labels.

HOW TO CHECK THE REFERENCE

In Chart_Build, set the vertical axis to Minimum 0 and Maximum 1, then compare the bar lengths with the Step 3 ratio.



KEY TERMS

Key terms

Truncated Axis:

A quantitative axis that starts above zero, distorting bar length.

Baseline:

The reference from which bar length is judged, normally zero.

Clutter:

Visual elements that consume attention without adding evidence.

Direct Label:

A value written on or beside the data mark instead of in a separate legend.

Non-Color Cue:

A pattern, outline, label or position that carries meaning without relying on color.

✓ STUDY CHECK Put the learning into action

- Choose a table, bar or line by the viewer's task and the statistic being shown.
- Apply the critique protocol: meaning and denominator first, then encoding and accessibility, then the title.
- Build charts from linked, checked cells with a zero baseline, direct labels, denominators and a non-color cue.

Bounded Data Stories and Independent Validation

CONCEPT From Exploratory Analysis to an Explanatory Story

Exploratory analysis searches across possible patterns, questions and data issues. **Explanatory communication** selects checked evidence for a defined audience and decision; purpose, not polish, is the difference. A **bounded story** says only what the records support, linking audience and decision, observed result, comparison, bounded action, limitation and next check. An **explanatory title** states the supported takeaway, and an **annotation** adds decision context rather than repeating a label.

CONCEPT How Claim Verbs and Validation Work

The strongest verb sets the evidence standard. "Observed" reports a value; "lower than" compares; "associated with" notes co-occurrence; "predicts" needs testing on new outcomes; "caused" needs evidence that changing one factor changed another after other explanations were ruled out. Four descriptive weeks support the first three. **Validation** challenges a result independently: reconcile counts, inspect records, recalculate by a second route and compare labels, cells and narrative. Re-entering the same formula checks arithmetic only. The **traceability chain** runs from decision to grain, KPI, statistic, chart, claim, action, limitation and next check.

**WORKED EXAMPLE****Worked Evidence: Validating the MetroMeal Story**

The session's model story: "For the operations director deciding whether to investigate weekend staffing before the next schedule, on-time delivery for August 24 through August 30 was 78.4%, or 392 of 500 eligible deliveries. This was 6.6 percentage points below the 85% target and lower than the prior three observed weeks, which ranged from 84.0% to 87.6%. The complaint rate remained within its 3% guardrail at 2.8%. Review order-level timing by day and relevant staffing records before reallocating resources because four descriptive weeks do not establish the cause or the likely effect of a staffing change." Each number is checked below.

WORKED EVIDENCE**Population:** Eligible completed MetroMeal deliveries**Period:** August 3 through August 30, 2026**Unit:** Story phrases and their checks

Story phrase	Source cell	Independent check	Result
392 of 500	KPI_Build!D10, C10	$392 + 108 - 500$	0
78.4%	KPI_Build!E10	$392 \div 500$ by hand; SPSS H5	0.784
6.6 points below	KPI_Build!F10	$85.0 - 78.4$	6.6
84.0% to 87.6%	KPI_Build!E7:E9	$420 \div 500$ and $438 \div 500$	84.0%, 87.6%
2.8%	KPI_Build!H10	$14 \div 500$ by hand	0.028

Interpretation: Every number traces to a checked cell and a second route, and the final sentence names the causal limit.

CONCEPT**Interpreting an Exploratory Clue Without Overclaiming**

The lab's optional Topic 2 splits weeks by weekend_flag. In the latest week, 320 of 358 weekday deliveries (89.4%) and 72 of 142 weekend deliveries (50.7%) were on time. Computed the same way from `orders_story.csv`, earlier weekend rates were 97 of 142 (68.3%), 102 of 142 (71.8%) and 97 of 142 (68.3%). This exploratory observation strengthens the case for investigating weekend operations, but

weather, restaurant delays, demand, routes and order mix may also differ on weekends. A bounded sentence: "In the week of August 24, the weekend on-time rate was 50.7% (72 of 142), compared with 89.4% on weekdays; weekend staffing is one hypothesis to investigate, not an established cause."

COMMON MISTAKE Common Misconception: A Real Miss Proves the Cause

The most common story error keeps a causal sentence because the rate really did miss the target. A real miss does not reveal its cause. "Weekend understaffing caused a delivery crisis" adds an undefined dramatic word to a causal verb; the session asks writers to remove "collapse", "crisis", "significant", "caused" and "will" unless supported. A row of Self_Check ticks is not proof either: it compares numbers and counts words, and cannot detect a causal claim. Avoid generic limits such as "More research is needed"; write "Four descriptive weeks do not identify the cause or expected effect of a staffing change."

LIMITS AND CONDITIONS Limits of a Bounded Story and the Bridge to Session 3

A bounded story is only as reliable as its data. If table, formula, chart and narrative disagree, stop and resolve the difference before recommending action. The session's forward-check list for Session 3 covers missing completion times, duplicate order identifiers, changed eligibility, unusual delays and whether mean, median, spread and shape agree. If missing records were disproportionately late, the true rate would be lower. Generalizing beyond the window needs later sampling methods, and a causal claim would need credible variation in staffing, aligned comparison periods, reliable scheduling records and checks for other explanations.

GUIDED PRACTICE Guided Practice: Is GreenBlock's Story Bounded?

GreenBlock is a made-up organization from Lab Part I. Over three observed months, 273 of 300 eligible repairs were completed within the service window, against a 95.0% target, and a draft claims a new policy caused the miss. Task 1: Calculate the rate and the gap in points. Task 2: Separate what the records show from what the draft claims. Task 3: Write a bounded title, an action (investigate before revising) and a limitation naming the three-month scope. Task 4: Write exactly three sentences. Check: 91.0% and -4.0 points.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

TRANSFER

Changed-Case Transfer: Two Audiences, One Set of Facts

Adapt the MetroMeal story for two readers. The executive version leads with the target miss and the investigation decision; the analyst version adds the 392 of 500 denominator, the KPI definition, source checks and the four-week limitation. A circulated document, unlike a live talk, must carry definitions, labels, denominators, window and limits on the page. List what changed between versions and what could not change: facts, units and causal boundaries.

Optional independent practice: my response

Write your approach, supporting evidence and assumptions here. After checking the explanation, record what you would revise.

CONCEPT

Reconciliation Identity

$$O + L - E = 0, \quad \frac{O}{E} + \frac{L}{E} = 1$$

SYMBOLS AND PRONUNCIATION

- O : on-time eligible deliveries; L : late eligible deliveries; E : eligible deliveries

INPUTS AND ASSUMPTIONS

Every eligible delivery is classified as either on time or late, never both and never neither.

Worked steps

- 1 Step 1: Substitute the latest week: $392 + 108 - 500 = 0$.
- 2 Step 2: Check the complements: $0.784 + 0.216 = 1$, so the on-time and late rates cover all eligible deliveries.
- 3 Step 3: Reach the late count a second way, by counting rows with late_minutes above 0: 108, then confirm KPI_Build!G18 shows 0.

WORKED RESULT

The identity holds: $392 + 108 = 500$; $78.4\% + 21.6\% = 100\%$.

WHAT THE RESULT MEANS

A nonzero result would signal missing, duplicated or misclassified records before any rate is reported.

WHEN TO USE IT

Before a number goes into a story, chart title or recommendation.

LIMITS AND CONDITIONS

Reconciliation confirms internal consistency. It cannot detect deliveries that never entered the file.

HOW TO CHECK THE REFERENCE

SPSS Block H5 computes reconcile = ontime + late – eligible and lists 0.



KEY TERMS

Key terms

Exploratory Analysis:

A broad search for possible patterns, questions and data issues.

Explanatory Communication:

Selected, checked evidence presented for a defined audience and decision.

Bounded Claim:

A statement limited to the observed population, period, comparison and design.

Causal Claim:

A statement that one factor produced an outcome; it requires evidence that isolates the proposed cause.

Independent Validation:

A challenge to a result through reconciliation, source inspection, a second calculation route or comparison across artifacts.



STUDY CHECK

Put the learning into action

- Draft a story with the five-part frame before formatting any chart.
- Trace every quantitative phrase to a checked cell and verify at least one number by a second route.
- Underline the strongest verb in each sentence and confirm that the evidence supports it.

**REFERENCE SUMMARY****Reference summary**

Use these concise reminders after reading the complete explanations above.

CONCEPT**Choose the Display****TASK DECIDES FORM**

- Table for exact lookup, bars for magnitude comparison, lines for ordered sequence.

MATCH THE STATISTIC

- A weekly rate is a proportion with a denominator; a target line is a benchmark, not a fitted value.

CONCEPT**Repair the Chart****FIVE REPAIRS**

- 0% to 100% bar axis, direct labels with n, 85.0% target line, non-color cue on the latest week, and a title quoting checked values.

WHY ZERO MATTERS

- On a 75% to 90% axis, 78.4% and 87.6% bars look 3.7 times different; on a zero axis they differ by about 12%.

CONCEPT**Write the Bounded Story****THREE SENTENCES**

- What the evidence shows; what it does not show; what the director should do next.

VERBS

- Use "observed", "lower than" and "warrants investigation". Avoid "caused", "collapse" and "will" without supporting evidence.

CONCEPT**Validate Independently****RECONCILE AND RECOMPUTE**

- $392 + 108 - 500 = 0$; $392 \div 500 = 78.4\%$; $85.0 - 78.4 = 6.6$ points; $14 \div 500 = 2.8\%$.

COMPARE ARTIFACTS

- Chart labels, table cells and story numbers must agree before the story is shared.

✓ STUDY CHECK

Study check: Honest Charts, Bounded Stories, and Validation

Use this ungraded check after reading. Close the explanation before responding, then reopen it to identify a specific correction.

1. Explain one central idea in your own words and connect it to a worked example above.
2. Reconstruct the example's inputs, sequence and interpretation; state what evidence supports each step.
3. Change one assumption or condition. Explain whether the same approach still applies.
4. Name a likely mistake and the check you would use to detect it.

My response

Write your explanation, evidence and changed-condition reasoning here.

MY READINESS

Choose one: **Can explain independently** / **Can explain with the reference**
/ Need to revisit.

Record the passage to revisit and the next action you will take.

Use your notes in the session readiness check below.

✍ CONNECT TO THE LAB

Connect this section to the lab

[Open Part G in the lab guide →](#)

Session lab reference

S02P (ungraded practice): Can you trace a business story from decision through KPI, grain, chart, claim, and validation?

Open the assigned task

Lab Parts G and H (hand-off at slide 70, return at slide 71), Lab Parts I and J (hand-off at slide 78, return at slide 79), then Lab Part M: Does the chart show the checked numbers honestly? and Lab Part N: What should the director do next? (guided build from slide 80)

Student lab file

[S02P_Learning_Workbook.xlsx](#) (card guide: [S02P_Student_Lab.html](#))

Workbook location

'Class_Practice'!B56:B88, 'Slide_Practice'!B56:B88, 'Chart_Build'!A4:E15 and 'Response'!B7, with checks in 'Self_Check'!B5:B29

SPSS location

Blocks H4 and H5, Guided Steps 3 and 4

Primary task

Build a linked, honest chart and write exactly three bounded sentences for the director.

Supporting task

Critique the RiverTown and MetroMeal charts, bound the GreenBlock and MetroMeal stories, and verify at least one number a second way.

Excel task

Link Chart_Build!A5:E8 to KPI_Build, insert the combo chart with a 0 to 1 axis, direct labels from E5:E8 and a non-color cue, then write Response!B7. Checks: B8 shows 78.4%, E8 shows 78.4% (n = 500), and every completed Self_Check row shows ✓.

SPSS task

Run Block H4 (zero-baseline bars) and Block H5 (reconcile = 0) at the hand-offs, then Guided Steps 3 and 4 during the guided build; optionally enter Excel and SPSS values side by side in Compare_Tools.

Bring back to the discussion

One statistical defect, one accessibility defect, one repair and a revised bounded title; then one explanatory title, a three-sentence bounded story, one action, one limitation and one next validation check.

Follow the lab's own prerequisites, instructions and submission rules. The Toolkit is a study reference and does not change those requirements.

My connection

Which idea from this section supports my lab decision, and what evidence will I check?

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Industry Practice

Use these explanations alongside Appendix C in the student slides. The figures are those reported in the session materials; the teaching inferences are labeled separately.

Slide 105: United States Department of Transportation

A cancellation rate and its reporting scope

- Reported: 16.3 million flights and a 2023 cancellation rate below 1.2%, fewer than 1.2 cancellations per 100 covered flights.
- Teaching inference: the rate covers the reported system for the year, not every airline, route or month, and it does not show that any policy caused it.

Source: Session 02 student slides, Appendix C, slide 105.

Slide 106: United Parcel Service

Scale is not service quality

- Reported: 91.1 billion US dollars in 2024 revenue, 5.7 billion delivered packages and about 22.4 million packages per day.
- Teaching inference: dollars, package counts and daily averages measure scale in different units; none shows on-time performance, and a daily average can hide peaks.

Source: Session 02 student slides, Appendix C, slide 106.

Slide 107: Netflix Advertising

Absolute change, relative change and the counted unit

- Reported: advertising-plan monthly active users rose from 40 million in May 2024 to 70 million in November 2024.
- Teaching inference: the increase is 30 million users, or $30 \div 40 = 75\%$ relative growth, not 30 percentage points; a monthly active user is not automatically a paying account.

Source: Session 02 student slides, Appendix C, slide 107.

Slide 108: Wells Fargo and the Consumer Financial Protection Bureau

Accounts, customers and amounts are different units

- Reported: over 2 billion US dollars in customer redress and a 1.7 billion US dollar civil penalty, covering more than 16 million accounts.
- Teaching inference: an account is not a unique customer or incident, and redress (to consumers) differs from a penalty (to the government), so redress per account can hide unequal harm.

Source: Session 02 student slides, Appendix C, slide 108.

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